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1. A particle A of mass $2m$ and a particle B of mass $3m$ are moving along the same straight line on a smooth horizontal surface. The particles are moving in opposite directions towards each other when they collide directly.

Immediately before the collision, the speed of A is $4u$ and the speed of B is ku .

Immediately after the collision, A rebounds with speed v and the speed of B is v .

The magnitude of the impulse received by A in the collision is $12mu$.

(a) Find v in terms of u only. [3]

(b) Find the two possible values of k . [5]



2. The first diagram shows two spheres, A and B , of equal radii on a smooth horizontal surface. Their masses are 1 kg and 3 kg respectively.

Take motion from left to right in each diagram to be positive.

Initially, sphere A travels at speed 2 m s^{-1} towards B , which is at rest. The spheres collide and the coefficient of restitution between A and B is e .

- (a) Show that, after the collision, the velocity of A is $\frac{1}{2}(1 - 3e) \text{ m s}^{-1}$, and find an expression for the velocity of B in terms of e . [4]
- (b) During the collision, the kinetic energy of the system decreases by 48%. Determine the value of e . [3]
- (c) Explain why the assumption that A and B have equal radii was needed in part (a). [1]



The second diagram shows two spheres, C and D , of equal radii on a smooth horizontal surface. Their masses are 2 kg and 5 kg respectively.

Sphere C is behind D and both move along the same straight line in the same direction, with C having speed $u \text{ m s}^{-1}$ and D having speed 1 m s^{-1} , where $u > 1$.

The spheres collide and during the collision C exerts an impulse on D of magnitude $\frac{10}{7}(u - 1) \text{ N s}$.

- (d) Show that C and D have the same velocity after the collision. [4]
- (e) Determine the fraction of kinetic energy lost due to the collision between C and D as $u \rightarrow \infty$. [3]

- 3.** Two trolleys, A and B , are modelled as particles of masses 0.5 kg and 1.0 kg respectively, moving along the same straight horizontal track.

Trolley A is travelling at 7 m s^{-1} and catches up with trolley B , which is travelling in the same direction at 2 m s^{-1} . The trolleys collide directly.

After the collision, trolley B moves at 5 m s^{-1} in the original direction of motion.

The coefficient of restitution between the trolleys is e .

- (a) Assuming that the track is smooth, show that $e = 0.8$. [5]
- (b) Describe one way in which the model could be refined. [1]

4. A particle P of mass 6 kg is projected vertically upwards from the ground with initial speed 15.8 m s^{-1}

At the same instant a particle Q of mass 3 kg is projected vertically downwards with speed 4.2 m s^{-1} from a point 20 m above the ground.

During the subsequent motion P and Q collide. The coefficient of restitution between P and Q is 0.5

Determine the time between this collision and P subsequently hitting the ground.

[10]

5. Two trolleys P and Q , of masses $2m$ and $5m$ respectively, move along the same straight line on a smooth horizontal track. Trolley P is behind trolley Q , and both are moving in the same direction. Trolley P catches up with trolley Q and they collide directly.

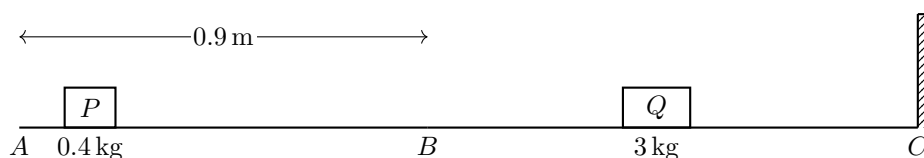
Immediately before the collision, the speed of P is ku and the speed of Q is u .

Immediately after the collision, trolley Q continues to move in the same direction with speed v and the speed of P is v .

The magnitude of the impulse received by Q in the collision is $10mu$.

(a) Find v in terms of u only. [3]

(b) Find the two possible values of k . [5]



6. The diagram shows two blocks P and Q of masses 0.4 kg and 3 kg respectively, on a horizontal surface. The points A , B and C lie on the surface in a straight line. The surface between A and B is rough and the surface between B and C is smooth. There is a wall at C .

The coefficient of friction between P and AB is $\frac{25}{49}$

Initially, P is at A and Q is at rest on BC . Block P is projected directly towards Q with speed 5 ms^{-1} . The two blocks collide on BC . As a result of the collision, P changes direction and comes to rest at A . You may assume that P only collides with Q once.

(a) Determine the coefficient of restitution between P and Q . [6]

(b) Calculate the impulse exerted on P by Q during the collision. [2]

After the collision with P , block Q strikes the wall at C . The collision between Q and the wall is perfectly elastic. After rebounding from the wall, Q comes to rest at a point on AB that is 0.30 m from A .

(c) Determine the coefficient of friction between Q and AB . [3]

7. Particle P has mass m and particle Q has mass $7m$.

The particles are moving in the same direction along the same straight line on a smooth horizontal surface. Particle P collides directly with particle Q .

Immediately before the collision, the speed of P is $9u$ and the speed of Q is u .

After the collision, the direction of motion of P is reversed.

The coefficient of restitution between P and Q is e .

- (a) Find the complete range of possible values of e . [7]

- (b) Given that $e = \frac{3}{4}$, find the total kinetic energy lost in the collision between P and Q . [4]

After the collision, P hits a smooth fixed vertical wall that is perpendicular to the direction of motion of P . Particle P rebounds.

The coefficient of restitution between P and the wall is f .

Given that there is a second collision between P and Q ,

- (c) find the complete range of possible values of f . [3]

8. A particle A of mass $3m$ is moving with speed $4u$ on a smooth horizontal plane towards a smooth fixed vertical wall. A particle B of mass m is on the same line between A and the wall, moving directly towards A with speed u .

Particles A and B collide directly. The coefficient of restitution between A and B is e , where $0 < e \leq 1$.

- (a) Show that immediately after the collision B moves towards the wall with speed $\frac{u}{4}(11 + 15e)$. [6]

- (b) Show that there will be a second collision between A and B after B has collided with the wall. [3]

The coefficient of restitution between B and the wall is $\frac{1}{3}$

Find, in simplified form, in terms of m , u and e ,

- (c) the magnitude of the impulse received by B from the wall, [3]
- (d) the loss in kinetic energy of B due to its collision with the wall. [3]

9. A particle P of mass m is moving in a straight line with speed $5u$ on a smooth horizontal plane. It collides directly with a particle Q of mass $4m$ that is initially at rest on the plane.

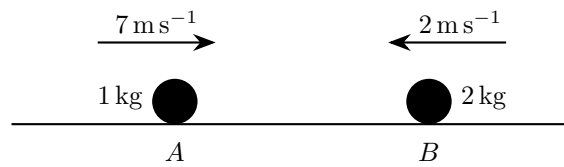
The coefficient of restitution between P and Q is e , where $e > \frac{2}{3}$.

- (a) Show that the speed of P immediately after the collision is

$$(4e - 1)u \quad [4]$$

After the collision, P moves in the opposite direction and strikes a smooth fixed vertical wall that is perpendicular to the direction of motion of P . The coefficient of restitution between P and the wall is f .

- (b) Find, in terms of e , the set of values of f for which there will be a second collision between P and Q . [6]



10. Two small uniform smooth spheres A and B have masses 1 kg and 2 kg respectively. The spheres are moving towards each other in the same straight line on a smooth horizontal surface. Sphere A is moving to the right with speed 7 m s^{-1} and sphere B is moving to the left with speed 2 m s^{-1} . The spheres collide. After the collision, A moves with speed 1 m s^{-1} .

Determine the possible speeds with which B moves after the collision.

[4]

- 11.** Particles P and Q , of masses m and $3m$ respectively, move on a smooth horizontal plane in the same straight line towards a vertical wall. Particle Q is in front of P . Initially, P has speed $6u$ and Q has speed u . Particle P catches up with Q and the particles collide before either particle reaches the wall.

The coefficient of restitution between the particles is e .

As a result of the collision, the direction of motion of P is reversed.

- (a) Find, in terms of u and e , the speed of P after the collision. [6]

After the collision, Q continues to the wall and rebounds. The coefficient of restitution between Q and the wall is $\frac{1}{3}$.

Given that there is a second collision between P and Q ,

- (b) find the full range of possible values of e . [5]

12. Two small uniform smooth spheres A and B , of masses 2 kg and 6 kg respectively, are on a smooth horizontal surface. Sphere A is projected directly towards B with speed 4 m s^{-1} . Sphere B is initially at rest. The coefficient of restitution between A and B is e .

(a) Show that the velocity of A after the collision, in the original direction of motion of A , is $(1 - 3e)\text{ m s}^{-1}$ and find a similar expression for the velocity of B . [5]

(b) The following three parts are independent of each other, and each considers a different scenario regarding the collision between A and B .

(i) In the collision between A and B the spheres coalesce to form a combined body C . State the speed of C after the collision. [1]

(ii) In the collision between A and B the direction of motion of A is reversed. Find the range of possible values of e . [2]

(iii) The total loss in kinetic energy due to the collision is 9 J . Determine the value of e . [4]